

ARM7-LPC2148-10 USER MANUAL

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CHAPTER 1: INTRODUCTION

The **ARM7-LPC2148-T10** board is a study Board which includes LPC2148 ARM7TDMI-S micro-controller with USB 2.0 Full speed device, multiple UART, SPI, I2C & on-chip 512K Flash and SRAM up to 40kB, produced by NXP semiconductors.

The ARM7TDMI-S is a general-purpose 32-bit microprocessor, which offers high performance and very low power consumption. The ARM processor is based on Reduced Instruction Set (RISC) architecture, and the instruction set and related decode mechanism are much simpler than those of micro programmed Complex Instruction Set Computers. This simplicity results in a high instruction throughput and impressive real-time interrupt response from a small and cost-effective processor Core.

Pipeline techniques are employed so that all parts of the processing and memory systems can operate continuously. Typically, while one instruction is being executed, its successor is being decoded, and a third instruction is being fetched from memory.

The ARM7TDMI-S processor also employs a unique architectural strategy known as THUMB, which makes it ideally suited to high-volume applications with memory restrictions, or applications where code density is an issue. The key idea behind THUMB is that of a super-reduced instruction set. Essentially, the ARM7TDMI-S processor has two Instruction sets:

- The standard **32-bit** ARM instruction set.
- A **16-bit** THUMB instruction set.

The THUMB set's 16-bit instruction length allows it to approach twice the density of standard ARM code while retaining most of the Arm's performance advantage over a traditional 16-bit processor using 16-bit registers. This is possible because THUMB code operates on the same 32-bit register set as ARM code. THUMB code is able to provide up to 65% of the code size of ARM, and 160% of the performance of an equivalent ARM Processor connected to a 16-bit memory system.

1.1 MICRO-CONTROLLER FEATURES

Following are the features of 16 bit /32 bit LPC2148 Arm Micro-controller

- **NXP LPC2148** is a 16-bit or 32-bit Micro-controller in a LQFP64-pin Package.
- 40kB of on-chip static RAM and 512kB of on-chip flash memory. 128-bit wide interface / accelerator enables high-speed 60 MHz operation.
- The LPC2148 provides 100000 erase / write cycles and 20 years of Data-retention.
- In-System Programming / In-Application Programming (ISP/IAP) via on-chip boot loader software. Single flash sector or full chip erase takes 400ms and Flash programming takes 1ms per 256-byte line.
- USB 2.0 Full speed compliant device controller with 2kB of endpoint RAM. In addition, the LPC2148 provides 8kB of on-chip RAM accessible to USB by DMA.
- Embedded ICE-RT and Embedded Trace Macro cell (**ETM**) interfaces offer real time debugging with on-chip Real Monitor software and high-speed real-time tracing of instruction execution.
- Two 10-bit ADCs provide a total of 14 analog inputs, with conversion times as low as 2.44µs per channel.

- Single 10-bit DAC provides variable analog output.
- Two 32-bit Timers / External event Counters (with four Capture & four Compare channels each), PWM unit (six outputs) and watchdog.
- Low power Real-Time Clock (RTC) with independent power and 32 kHz clock input.
- Multiple serial interfaces including two UARTs (16C550 equivalent), two Fast I2C-bus(400 kbit), SPI and SSP with buffering and variable data length capabilities.
- Vectored interrupt controller (VIC) with configurable priorities and vector addresses. Up to 45 numbers of 5 V tolerant fast general purpose I/O pins in a tiny LQFP64 package.
- Up to nine edge or level sensitive external interrupt pins available.
- 60 MHz maximum CPU clock available from programmable on-chip PLL with settling time of 100 μ s.
- On-chip integrated oscillator operates with an external crystal in range from 1 MHz to 30 MHz and with an external oscillator up to 50 MHz.
- Power saving modes include Idle and Power-down.
- Individual power enable/disable of peripheral functions as well as peripheral clock scaling for additional power optimization.
- Processor wake-up from Power-down mode via external interrupt, USB, Brown-Out Detect (BOD) or Real-Time Clock (RTC).
- Single power supply chip with Power-On Reset (POR) and BOD circuits: CPU operating voltage range of 3.0 V to 3.6 V ($3.3 \text{ V} \pm 10 \%$) with 5 V tolerant I/O pads.

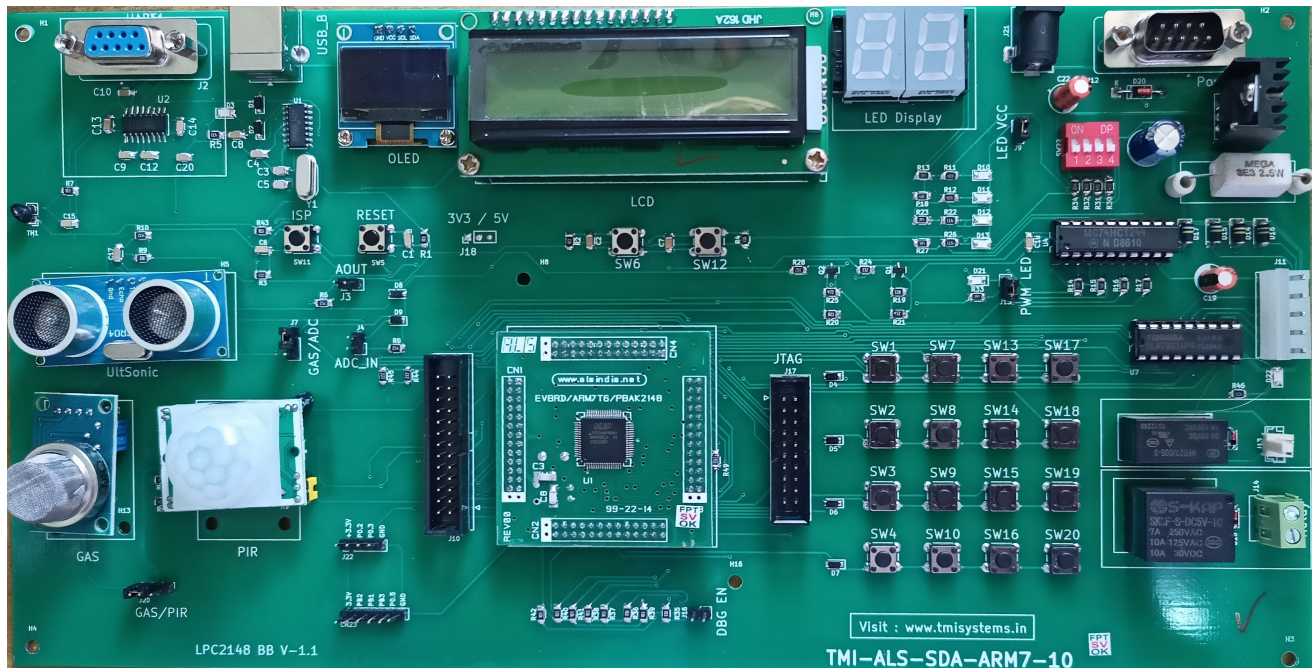
1.2 BOARD SPECIFICATIONS FOR LPC ARM 2148 EVALUATION BOARD TMI- SDA-ARM7-10

- LPC2148 16/32 BIT ARM7TDMI-S with 512K bytes Program Flash, 42K bytes RAM
- 12Mhz Crystal allows easy communication setup
- One on board voltage regulator for generating 3.3V. Input to this will be from External +5V DC
- Power- supply for external power supply 9 PIN D type and also provision for adaptor
- USB "B" type connector for downloading the program
- Piggy Back module containing LPC2148 controller
- Standard JTAG connector with ARM 2X10 pin layout for programming / debugging with ARM – JTAG
- Standard 26 pin FRC Connectors to connect to on-board interface or some of ALS Standard external interfaces
- Reset one push button switch for resetting the controller
- I2C & SPI Port are terminated in Berg for external interfacing

- 1 channel Internal 10 bit ADC
- On chip 8-bit DAC interface circuit
- UART0- USB connector for serial communication & UART1 9 pin DSUB for user application
- PWM One PWM channel is interfaced to LED with test point to observe the PWM output.
- Two external interrupt circuits
- Onboard 1" OLED
- 16X2 alphanumeric LCD
- 4x4 key matrix connected to the port lines of the controller
- Four general purpose LED's
- Stepper motor interface with direction and speed control
- DC motor interface with direction and speed control
- **On board Sensors like**
 - TEMPERATURE
 - PIR SENSOR
 - GAS SENSOR
 - ULTRASONIC Sensor
- Compact elegant plastic enclosure
- Operates off 5V DC

CHAPTER 2: HARDWARE DETAILS

SECTION 2.1: MAIN CONTROLLER BOARD (TMI-SDA-ARM7-10)



2.1.1 CONNECTORS DETAILS:

J1A1 CONNECTOR: 28 pin 14 X 2 HEADER is connected to the controller.

PIN #	DESCRIPTION	PIN #	DESCRIPTION
1	TDO-JTAG	2	TDI-JTAG
3	TMS-JTAG	4	TRST-JTAG
5	TCK-JTAG	6	P0.11 (PC3)
7	P0.10 (PC2)	8	P0.9 (PC1)
9	P0.8 (PC0)	10	3.3V
11	GND	12	3.3V
13	NC	14	NC
15	GND	16	NC
17	RESET	18	NC
19	3.3V	20	P0.25
21	NC	22	NC
23	NC	24	NC
25	NC	26	P1.24 (2148)
27	3.3V	28	NC

J2A1 CONNECTOR: 28 pin 14 X 2 HEADER is connected to the controller.

PIN #	DESCRIPTION	PIN#	DESCRIPTION
1	NC	2	NC
3	NC	4	P0.8

PIN #	DESCRIPTION	PIN#	DESCRIPTION
5	3.3V	6	NC
7	NC	8	GND
9	NC	10	NC
11	P1.20	12	P1.21
13	P1.22	14	P1.23
15	NC	16	NC
17	NC	18	GND
19	3.3V	20	NC
21	NC	22	NC
23	NC	24	NC
25	P0.22	26	P0.23
27	P0NC	28	NC

J3A1 CONNECTOR: 28 pin 14 X 2 HEADER is connected to the controller.

PIN #	DESCRIPTION	PIN #	DESCRIPTION
1	NC	2	NC
3	P1.25	4	NC
5	NC	6	3.3V
7	GND	8	P0.31 (P1.24) (PA7)
9	P0.30 (PA6)	10	P0.22 (PA5)
11	P0.28 (PA4)	12	P1.19 (PA3)
13	P1.18 (PA2)	14	P1.16 (PA0)
15	P1.17 (PA1)	16	P0.7
17	P0.6	18	P0.5
19	P0.4	20	P0.3
21	P0.2	22	P0.15 (PC7)
23	3.3V	24	GND
25	P0.14 (PC6)	26	P0.13 (PC5)
27	P0.12 (PC4)	28	NC

J4A1 CONNECTOR: 28 pin 14 X 2 HEADER is connected to the controller.

PIN #	DESCRIPTION	PIN #	DESCRIPTION
1	P0.21 (PB5)	2	P0.20 (PB4)
3	P0.19 (PB3)	4	P0.18 (PB2)
5	P0.17 (PB1)	6	P0.16 (PB0)
7	NC	8	GND
9	3.3V	10	NC
11	NC	12	NC
13	NC	14	NC
15	NC	16	NC
17	NC	18	NC
19	NC	20	NC
21	3.3V	22	GND
23	TXD0 (P0.0)	24	RXD0 (P0.1)
25	RTCK-JTAG	26	NC
27	NC	28	NC

J17 CONNECTOR:

20 pin FRC connected to the controller, Standard JTAG connector for programming/debugging with ARM-JTAG debugger. SHORT jumper JP1 for JTAG to work.

PIN #	DESCRIPTION	PIN #	DESCRIPTION
1	+3.3V	2	+3.3V
3	TRST	4	GND
5	TDI	6	GND
7	TMS	8	GND
9	TCK	10	GND
11	RTCK	12	GND
13	TDO	14	GND
15	RST	16	GND
17	GND	18	GND
19	GND	20	GND

J10 CONNECTOR: 26 pin FRC connected to the controller, which is compatible with **ALS Standard External Interfaces**.

PIN #	DESCRIPTION	PIN #	DESCRIPTION
1	P0.12 (PC4)	2	P0.13 (PC5)
3	P0.10 (PC2)	4	P0.11 (PC3)
5	P0.8 (PC0)	6	P0.9 (PC1)
7	P0.22 (PB6)	8	P0.23 (PB7)
9	P0.20 (PB4)	10	P0.21 (PB5)
11	P0.18 (PB2)	12	P0.19 (PB3)
13	P0.16 (PB0)	14	P0.17 (PB1)
15	P0.30 (PA6)	16	P0.31 OR P1.24 (PA7)
17	P0.28 (PA4)	18	P0.29 (PA5)
19	P1.18 (PA2)	20	P1.19 (PA3)
21	P1.16 (PA0)	22	P1.17 (PA1)
23	P0.14 (PC6)	24	P0.15 (PC7)
25	+5V	26	GND

2.1.2 CONNECTOR DETAILS:
J12 CONNECTOR:

9-Pin D-type Male Power connector.

Pin Number	Description
4, 5	GND
9	+5V

J1 USB "B" type CONNECTOR:

UART0 USB B-type Female connector connects to the serial port of host computer for In-System Programming (ISP) application and transferring the data between controller device and host computer.

Pin Number	Description
1	Vbus
2	D-
3	D+
4, 5 (Shield)	GND

J2 CONNECTOR:

(UART1) 9-Pin D-type Female connector is used for user application.

Pin Number	Description
2	CONTROLLER Rx1
3	CONTROLLER Tx1
5	GND

2.1.3 JUMPER and CONNECTOR DETAILS:

JUMPERS	CONNECTION	DESCRIPTION
J1		USB "B" type connector for ISP programming
J2		9 pin D SUB connector for user application
J3		AOUT (DAC) output
J4		Input to internal ADC
J5 (1,2)	Closed	ADC Input is connected to P0.25 pin of Controller
J5 (2,3)	Closed	PIR sensor Output is connected to P0.25 pin of Controller
J6		Ultrasonic Connector
J7 (1,2)	Closed	It connects to the internal ADC
J7 (2,3)	Closed	It connects to GAS Sensor
J8		OLED connector
J9		Connects Vcc to the general purpose LED's
J10		26 pin Connector for external interface
J11		Stepper motor connector
J12		9 pin "D" sub connector for Power supply.
J13		DC Motor connector
J14		Relay O/p connector
J15	Closed	PWM Output
J16 (1,2)	Closed	Enable JTAG Programming.
J17		JTAG connector
J18 (1,2)	Closed	Enable 3.3V to OLD circuit.
J18 (2,3)	Closed	Enable Vcc to 16×2 LCD circuit.
J19		PIR sensor Connector
J20 (1,2)	Closed	Connects to PIR sensor
J20 (2,3)	Closed	Connects to GAS sensor
J21		Adapter output to Power supply input to the trainer
J22		I2C pin out connector
J23		LCD Connector
CN23		SPI Connector

2.1.6 IC DETAILS

U1	CH340G USB Interface Integrated Circuit
U2	MAX3232 RS232 LINE DRIVER(16 PIN)
U4	74HCT244 OCTAL BUFFER (20 PIN)
U5, U6	LT543 SEVEN SEGMENT DISPLAY (10 PIN)
U7	ULN2803 DRIVER 8 Darlington array(18 PIN)

2.1.7 SWITCH DETAILS:

SW5 - Reset Switch.
SW11 - ISP Switch. line for ISP. During programming the switch has to be kept short.
SW6 - External interrupt to P1.25 pin of Controller.
SW12 - External Interrupt connected to P1.24 of Controller.
SW1 to SW4,
SW7 – SW10,
SW13 – SW20 – 4×4 key matrix.
SW22 - 4 way DIP switch for general purpose.

2.1.8 J11 STEPPER MOTOR POWER-MATE DETAILS:

5 Pin Power mate (High Current Output Lines for Stepper Motor).

Pin Number	Description
1	+5V
2	OUT1(A)
3	OUT2(B)
4	OUT3(C)
5	OUT4(D)

2.1.9 J23: 4 pin Single female Berg for LCD Interface.

PIN #	DESCRIPTION
1	GND
2	+5V
3	P0.2 (SCL0)
4	P0.3 (SDA0)

J4: 2 pin reliamate male analog input for internal ADC.

Pin Number	Description
1	Input signal
2	GND

J3: ANALOG Output Interface

Pin Number	Description
1	Analog O/P
2	GND

J13: DC MOTOR O/p

Pin Number	Description
1	POLE1 of Rly1
2	POLE2 of Rly1

J14: Relay output

Pin Number	Description
1	Comm of Rly2
2	NO of Rly2

CHAPTER 3: CABLE DETAILS

3.1 APPLYING POWER:

Use the following procedure to apply power. Connect a 5V adapter to the trainer kit marked as J21, or Connect a 9-pin DSUB Female Connector to a 9-pin DSUB Male connector **J12** provided on the Board. The color code for the supply is shown in table below:

PIN #	POWER	COLOR CODE
9	+5V	BLUE
4,5	GND	YELLOW / BLACK

3.2 SERIAL COMMUNICATION:

The USB "B" type cable is required for communicate between the trainer kit and host computer. Open Flash Magic & set device, baud rate to **38400**, Comm port as shown in the device manager, Oscillator as **12** MHz.

CHAPTER 4: INSTALLATION

4.1 Keil u Vision 4 ID INSTALLATION:

Installation of keilUVision4 as follows.

- Go to **EXE** folder and then **uvision4.2** in the CD and run **MDK474.exe** file.
- **Next**
- Click on the option "I agree to all the terms of..." and then give **Next**
- **Next**
- Give name and the mail id (it might be any mail id) and then **Next**
- Click **Finish** to complete the installation.

4.2 PROJECT CREATION IN KEILUV4 IDE:

Create a project folder before creating NEW project.

- Open **Keil uVision4 IDE** software by double clicking on "Keil Uvision4" icon.
- Go to "**Project**" then to "**New u Vision Project**" and save it with a name in the respective project folder, already you created.
- Select the device as "**NXP**" In that "**LPC2148**" then press **OK** and then press "**YES**" button to add "**startup.s**" file.
- In **startup** file go to **Configuration Wizard**. In **Configuration Wizard** window unchecked **PLL Setup** and check **VPBDIV Setup**.
- Go to "**File**" In that "**New**" to open an editor window. Create your source file and use the header file "**lpc21xx.h**" in the source file and save the file. **Colour syntax highlighting** will be enabled once the file is saved with a extension such as ".C".
- Right click on "**Source Group 1**" and select the option "**Add Existing Files to Group**
- **Source Group 1**" add the *.C source file(s) to the group.

After adding the source file you can see the file in Project Window.

Then go to "**Project**" in that "**Translate**" to compile the File (s). Check out the **Build output** window.

Right click on **Target1** and select **options for Target Target1**.

Then go to option "Target" in that

- Xtal 12.0MHz
- Select "Use Micro Lib".
- Select IROM1 (starting 0x0 size 0x80000).
- Select IRAM1 (starting 0x40000000 size 0x8000).

Then go to option "Output"

- Select "Create Hex file"

Then go to option "Linker"

- Select "Use Memory Layout for Target Dialog".

To come out of this window press **OK**.

Go to "**Project**" in that "**Build Target**" for building all source files such as ".C", ".ASM", ".h", files, etc... This will create the *.HEX file if no warnings & no Errors. Check out the **Build output** window.

4.3 FLASH MAGIC VERSION 6.1: Installation of Flash Magic as follows.

Go to **EXE** folder and then **Flash Magic 6.1** folder in the CD & run **FlashMagic.exe**

- Click on the option "I Accept the Agreement" and then give **Next**
- Then it asks the **Destination** location, Click **Next**.
- Further Select start menu folder, Click **Next**.
- Select "**Create a desktop icon**" then **Next**
- It prompts "**Ready to Install**" Click **INSTALL**.
- Click **Finish** to complete the installation.

4.4 ISP PROGRAMMING:

FLASH MAGIC software can be used to download the HEX files to the Flash memory of controller.

How to Download?

Switch ON the power. Connect USB "B" type cable between trainer kit and PC COM port. Push both RESET key and the ISP KEY at time and release RESET key first and release ISP key next.

Settings in FLASH MAGIC:

Options -> Advanced options -> Hardware Config

Enable these options only

- Use DTR and RTS to control RST and ISP pin
- Keep RTS asserted while COM port open Press OK then do the below settings

Step1. Communications:

- Device : LPC2148
- Com Port : COM1(Check and connect)
- Baud Rate : 38400
- Interface : None(ISP)
- Oscillator : 12MHz

Step2. ERASE:

- Select "Erase Blocks Used By Hex File".

Step3. Hex file:

- Browse and select the Hex file which you want to download.

Step4. Options

- Select "Verify after programming".

Step5. Start:

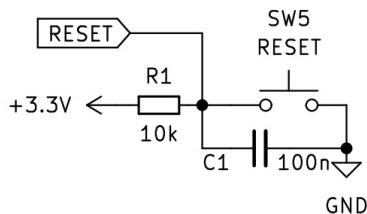
- Click Start to download the hex file to the controller.

After downloading the code press the RESET Key and program starts executing in the hardware.

CHAPTER 5: ONBOARD INTERFACES

5.1 Push Button Switch (RESET)

SW5: This push button is connected to the RST pin (57) of the Micro controller. It is used to reset the controller.



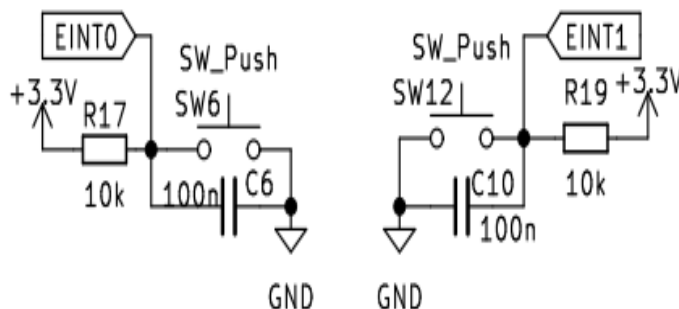
5.2 16X2 LCD Interface:

A 16X2 Alphanumeric LCD Display with back light is provided along with the Evaluation Board. The LCD is interfaced using I2C mode (HW610).

5.3 External Interrupt:

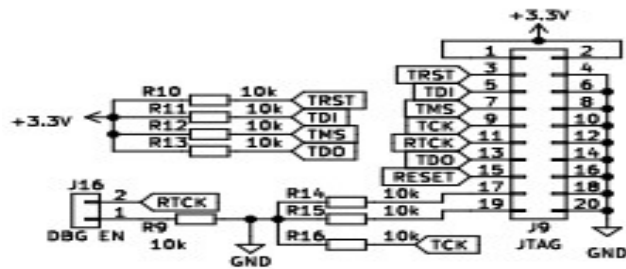
External Interrupt 0: By using P0.16 port line we are generating external interrupt (EINT0). Short JP6, when we press the switch SW2 the port line goes low & the external interrupt occurs at port line P0.16.

External Interrupt 1: By using P0.3 port line we are generating external interrupt (EINT1). Short JP12, when we press the switch SW3 the port line goes low & the external interrupt occurs at port line P0.3.



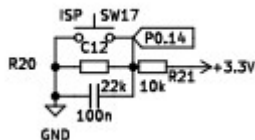
5.4 JTAG:

The JTAG connector allows the software debugger to talk via a JTAG (**Joint Test Action Group**) port directly to the core. Instructions may be inserted and executed by the core thus allowing LPC2148 memory to be programmed with code and executed step by step by the host software. The debug communication channel allows the JTAG port to be used for sending and receiving data without affecting the normal program flow. Short the jumper JP1 (1, 2) in the target board to enable JTAG programming.



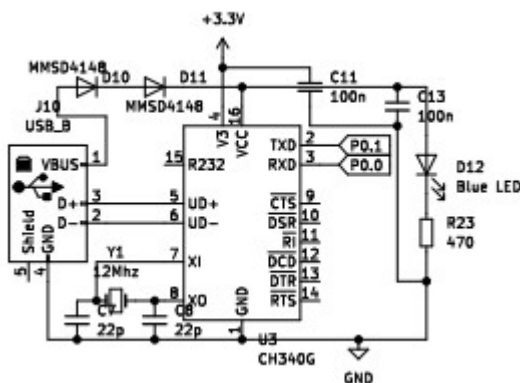
5.5 ISP Circuit:

A LOW level after reset at pin P0.14 is considered an external hardware request to start the ISP command handler. Assuming that power supply pins are on their nominal levels when the rising edge on RESET pin is generated, it may take up to 3 ms before P0.14 is sampled and the decision on whether to continue with user code or ISP handler is made.



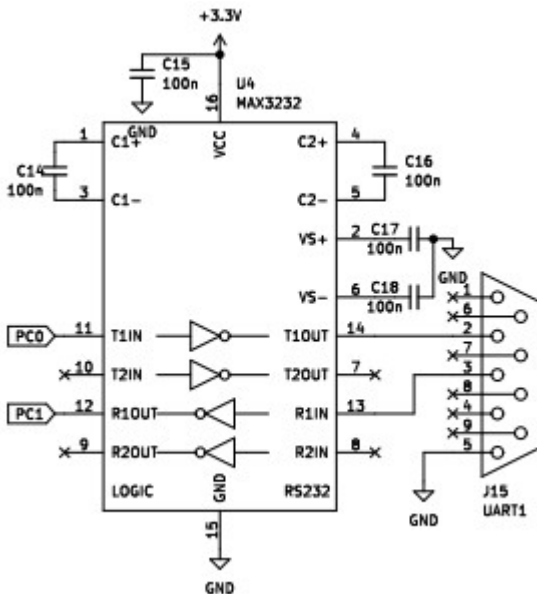
5.6 UART0:

The board has an USB to serial communication port. The D+ and D- lines are converted to transmit and receive lines using IC **CH340** USB to serial converter. Use USB "B" type cable to connect.



5.7 UART1:

The board has an RS-232 serial communication port. The RS-232 transmits and receives signals that appear on the female 9-pin DB connectors (DB3). Use a standard RS-232 cross cable to connect the board to the computer's serial port. The controller U1 provides serial I/O data at TTL levels to the MAX232 (U13) device, which in turn converts the logic value to the appropriate RS-232 voltage level.

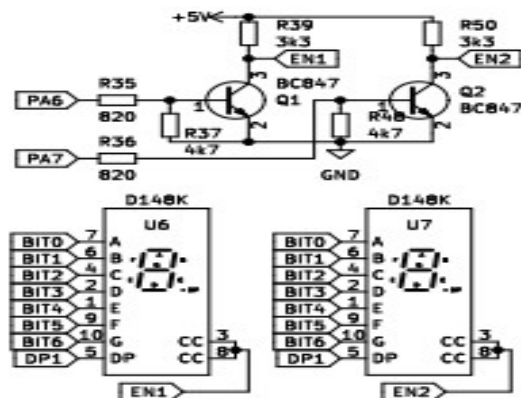


5.8 Two Digit Multiplexed 7- Segment Displays:

There are two multiplexed 7-segment displays LT543 (**U6 and U7**) on the board. Each display has 8-inputs SEG_A (Pin-7), SEG_B (Pin-6), SEG_C (Pin-4), SEG_D (Pin-2), SEG_E (Pin-1), SEG_F (Pin-9), SEG_G (Pin-10) and DP (Pin-5). The remaining pins pin-3 & 8 is Common Cathode. The port lines P0.30 and P0.31 are used to select one of the 2 digits as shown in the table below. The port lines P0.16 to P0.23 are used as segment lines for the EIGHT digits through the 74HCT244 buffer (**U10**).

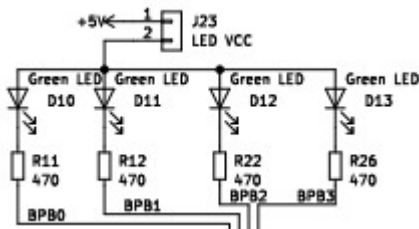
Selection Of seven segment displays:

P0.30	P0.31	Display unit selected
1	0	U6
0	1	U7



5.9 Flashing LED's:

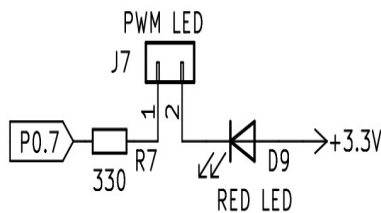
Light Emitting Diodes (LED's) are components most commonly used for displaying the port line status. There are 4 LEDs on the board; these lines are connected to the Port lines P0.16 (PB0) to P0.19 (PB3) through buffer.



5.10 PWM:

Using P0.8 (**PWM4**) port line we are testing PWM. Here match register (PWMMR4) is used to adjust the duty cycle (T-on to T-off ratio). Initially the duty cycle goes on increasing up to some specified value, when it reaches maximum value the duty cycle goes on decreasing and this process continues. Observe the output on CRO at TP8 and also LED (L11) will change status slowly.

Corresponding DC voltage can also be monitored at TP3



5.11 Stepper motor Interface:

The Stepper motor can be interfaced to the board by connecting it into the Power Mate J11. It is interfaced through the high current driver ULN2803. These lines will have high current (max 300 mA) with low voltage level of 0.7V. The rotating direction of the stepper motor can be changed through software. Port lines used for Stepper motor are P0.12 – P0.15.

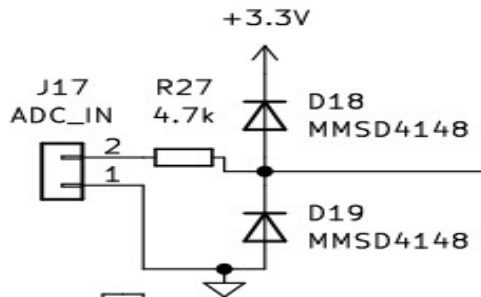
5.12 DC motor Interface:

The **DC Motor** can be interfaced to the board by connecting it to the Reliamate J5. It is interfaced through the high current driver ULN2803. These lines will have high current (max 300 mA) with low voltage level of 0.7V. The direction of the rotation can be changed through software using Relay **RLY1**. Port lines used for DC motor are P0.8 and P0.9.

5.16 Internal ADC interface:

On board there is one interface for internal ADC. Pin P0.25 of controller is used to convert the analog input voltage varied using that is given through J17/2. J17/1 is connected to GND. A 0.00 to 3.3V is the input voltage range. 000 to 3FF is the converted digital voltage range here.

Note: Make sure that one should not feed an I/p beyond 3.3V maximum.



5.17 GAS SENSOR:

Gas sensor is one of the features in this trainer. MQ2 is the sensor used in the trainer. The Grove – Gas Sensor(MQ2) module is useful for gas leakage detection (home and industry). It is suitable for detecting H₂, LPG, CH₄, CO, Alcohol, Smoke or Propane. Due to its high sensitivity and fast response time, measurement can be taken as soon as possible. The MQ-2 type smoke sensor is made of a tin dioxide semiconductor gas sensing material. When it works at a temperature of 200 to 300 °C, the tin dioxide adsorbs the oxygen in the air, which reduces the density of electrons on the semiconductor and thus increasing the resistance. The MQ-2 is a smoke and combustible gas sensor from Winsen. It can detect flammable gas in a range of 300 -10000ppm. It's most common use is domestic gas leakage alarms and detectors with a high sensitivity to propane and smoke. The O/p of the sensor is connected to pin P0.25 and P1.24 of the controller. The O/p of the sensor converted ADC value is sent to serial port.

5.17 ULTRASONIC SENSOR:

Ultrasonic sensor is other features of this trainer. HC-SR04 is the sensor used in the trainer. The HC-SR04 ultrasonic sensor uses sonar to determine the distance to an object. This sensor reads from 2cm to 400cm (0.8inch to 157inch) with an accuracy of 0.3cm (0.1inches), which is good for most hobbyist projects. In addition, this particular module comes with ultrasonic transmitter and receiver modules. Most ultrasonic sensors are based on the principle of measuring the propagation time of sound between send and receive (proximity switch). The barrier principle determines the distance from the sensor to the reflector (retro-reflective sensor) or to an object (through-beam sensor) in the measuring range. The O/p of the sensor is connected to pin P0.25. The O/p of the sensor ADC value is sent to serial port.

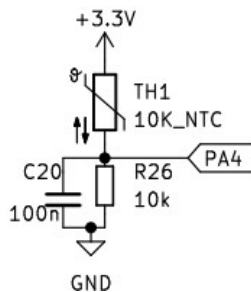
5.18 PIR SENSOR:

PIR Motion sensor HC-SR501 used in the trainer. This motion sensor module uses the LHI778 Passive Infrared Sensor and the BISS0001 IC to control how motion is detected. The module features adjustable sensitivity that allows for a motion detection range from 3 meters to 7 meters, also includes time delay adjustments and trigger selection that allow for fine tuning within your application. The SR501 will detect infrared changes and if interpreted as motion, will

set its output low. What is or is not interpreted as motion is largely dependent on user settings and adjustments. The device requires nearly a minute to initialize. During this period, it can and often will output false detection signals. Circuit or controller logic needs to take this initialization period into consideration. Time Delay Adjust Sets how long the output remains high after detecting motion anywhere from 5 seconds to 5minutes. Sensitivity Adjust Sets the detection range from 3 meters to 7 meters Trigger Selection Jumper Set for single or repeatable triggers. Output Pin Low when no motion is detected. High when motion is detected. High is 3.3V. The O/p of the sensor is connected to pin P1.24. The O/p of the sensor is sent to serial port.

5.19 Temperature SENSOR:

Temperature sensor NTC-10K 5MM is the sensor used in the trainer. It measure surface temperature with small gradients (difference between surface and NTC element temperature) due to low thermal conductivity of the nickel wire. NTC is a rugged product developed to withstand temperatures as high as 150 °C during thousands of hours without any wear out sign (no insulation resistance loss). The relatively large negative slope means that even small temperature changes cause a significant change in electrical resistance which makes the NTC sensor ideal for accurate temperature measurement and control. The resistance at 25 °C (substantially at room temperature) provides a convenient reference point for thermistors. Tolerances on R are due mainly to variations in ceramic material manufacture and tolerances on chip dimensions. Through the use of highly homogeneous material compositions and proprietary ceramic sawing techniques allowing precise control of chip dimensions, products are available with tolerances on R lower than 1%. The O/p of the sensor is connected to pin P0.28. The O/p of the sensor ADC value is sent to serial port.



CHAPTER 6:

DEMO PROGRAMS IN KEIL μ VISION4 μ VISION4 IDE:

For all the demo programs make sure that the corresponding settings have to be made:

- Press RESET switch(SW5) along with ISP switch(SW11) release SW5 and then SW11 for ISP programming.
- Use 26 pin connector for **External NIFC's**. Use 26 core 2 feet flat cable

6.1 TO test 7 Segment Display:

Load the counter.hex file using programming tool FLASH MAGIC.

Folder Name: 1 Counter

path: SOFTWARES\1 Counter\counter.hex

Settings: Remove the jumper J9

Result:

After downloading the *.HEX file, press RESET Switch SW5. Observe display changing from **00 to FF**. **Ex: 00, 11, 22, 33** etc up to **FF** and then back to **00**. This process is continuous in loop.

6.2 To test LED's:

Load the BinDisp.hex file using programming tool FLASH MAGIC.

folder name: 2 Binary_Dis

path: SOFTWARES\2 Binary_Dis\BinDisp.hex

Settings: Short jumper J9

Result:

Observe the toggling of LED's D10 – D13 in binary pattern 0 – F.

6.3 To test flashing LED's:

Load the blink.hex file using programming tool FLASH MAGIC.

folder name: 3LED_BLINK

path: SOFTWARES\3 Led_Blink\Blink.hex

Settings: short jumper J9

Result: observe the toggling of LED's D10 to D13. At a time all the LED's toggle.

6.4 TO TEST Pulse Width Modulation (PWM):

Load the pwm.hex file using programming tool FLASH MAGIC.

folder name: 4 PWM_LED

path: SOFTWARES\4 PWM_LED\Objects\PWM.hex

Settings: J15 jumper has to be shorted.

Result:

Observe the D21 intensity which varies according to PWM output. Open J15 jumper and check the PWM output waveform on CRO at J15/1 with respect to GND or check the corresponding DC voltage at J15/1 with respect to GND.

6.5 To test stepper motor

Load the STPM.hex file using programming tool FLASH MAGIC.

folder name: 5 Stepper motor

path: SOFTWARES\5 Stepper motor\STPM.hex

Connect the Female Power-mate of the stepper motor to the male Power-mate J11 present on the board.

Result:

The stepper motor rotates in **Clockwise & Anti Clockwise** direction. This process is continuous in loop.

6.6 To test DC motor

Load the dcmotor.hex file using programming tool FLASH MAGIC.

folder name: 6 DCMotor

path: SOFTWARES\6 DCMotor\dcmotor.hex

Connect the Female relimate of the DC motor to the male Relimate J13 present on the board.

Result: The DC motor rotates **Clockwise** with a delay. This process is continuous in loop.

6.7 To test Relay

Load the Relay.hex file using programming tool FLASH MAGIC.

folder name: 7 Relay

path: SOFTWARES\7 Relay\Relay.hex

Result: The Relay K1 is turned on and turned off with a regular interval. When the relay is on check to see both the pins of J14 is shorted with the help of a meter. Also observe the Led D22.

6.8 TO TEST Keypad (4X4 Matrix):

Load the KeyMatrix.hex file using programming tool FLASH MAGIC.

folder name: 8 KeyMatrix

path: SOFTWARES\8 KeyMatrix\KeyMatrix.hex

Result: When keys SW1 to SW20 are pressed the corresponding outputs '0 to F' will be displayed on the serial.

6.9 External interrupt0 Interface

Load the EXT_INT0.hex file using programming tool FLASH MAGIC.

folder name: 9 EXT_INT0

path: SOFTWARES\9 EXT_INT0\EXT_INT0.hex

Settings: short jumper J9

Result: When switch SW6 is pressed, the port line goes low & the external interrupt occurs at port line P0.16. To show the external interrupt has occurred LED D10 has been used. LED D10 toggles at each time when you press the switch SW6.

6.10 TEST INTERNAL_DAC

A) Ramp Wave:

Load the DAC.hex file using programming tool FLASH MAGIC.

File name: SOFTWARES\10 INT_DAC\ramp\Objects\DAC.hex

Download the file DAC.hex

Result: Press the reset switch to run the program. Observe the Analog output waveform at the Pin-1 of J3 using Oscilloscope (CRO) with respect to GND.

B) Sine Wave:

Load the sinewave.hex file using programming tool FLASH MAGIC.

File name: SOFTWARES\10 INT_DAC\sinewave\sinewave.hex

Download the sinewave.hex file.

Result Press the reset switch to run the program. Observe the Analog output waveform at the Pin-1 of J3 using Oscilloscope (CRO) with respect to GND.

C) Square Wave:

Load the DAC.hex file using programming tool FLASH MAGIC.

File name: SOFTWARES\10 INT_DAC\square\square.hex

Download the file DAC.hex

Result Press the reset switch to run the program. Observe the Analog output waveform at the Pin-1 of J3 using Oscilloscope (CRO) with respect to GND.

D) Triangle Wave:

Load the triangular.hex file using programming tool FLASH MAGIC.

File name: SOFTWARES\10 INT_DAC\triangular\triangular.hex

Download the triangular.hex file.

Result Press the reset switch to run the program. Observe the Analog output waveform at the Pin-1 of J3 using Oscilloscope (CRO) with respect to GND.

6.11 To TEST Internal ADC

Load the intADC.hex file using programming tool FLASH MAGIC.

File Name: SOFTWARES\11 IntADC\intADC.hex

Download the intADC.hex file.

Result: This routine is used to study the internal Feature of the LPC2148 Micro-controller. J4/1 is GND, wire a 10k pot externally. Connect the adjust pin of 10K pot to J4/2 and one end of the 10K pot is connected to 3.3V and other end of the pot is connected to GND. Observe the corresponding digital output value on serial. The input can be varied from 0.00V to 3.3V, the output displays from 000 to 3ff.)

6.12 To test LCD:

Load the LCD.hex file using programming tool FLASH MAGIC.

folder name: LCD

path: SOFTWARES\LCD\lcd.hex

Result:

****TMI-ALS ****

HELLO WORLD

The above message will be displayed on 16×2 LCD

6.13 To Test OLED

Load the OLED.hex file using programming tool FLASH MAGIC.

path: SOFTWARES\13 oled\Objects\OLED.hex

Result:

All the characters will be displayed on OLED display

6.14 TO TEST PIR Motion Sensor

Load the PIR.hex file using programming tool FLASH MAGIC.

folder name: 14 PIR

path: SOFTWARES\14 PIR\PIR.hex

Setting: Short J20/2-3 (at J20 short jumper pins marked as PIR)

Result: After downloading the program, open the Terminal window from Tools wizard in Flash Magic. Set the com port, baud rate as 9600 and press ok. Reset the trainer kit, when any movement near the PIR Sensor, **"Intruder Detected"** will be displayed on the terminal. This will be in continues loop.

6.15 TO TEST GAS Sensor

Load the gas.hex file using programming tool FLASH MAGIC.

folder name: 15 GAS

path: SOFTWARES\15 Gas\gas.hex

Setting: Short J20/1-2 (at J20 short jumper pins marked as GAS)

Result: After downloading the program, open the Terminal window from Tools Wizard in Flash Magic. Set the com port, baud rate as 9600 and press OK. Reset the trainer kit, the output of the Gas sensor will be displayed on the terminal as **"GAS OUT CH4= ***"** where *** is the value of the gas sensor. This will be in continues loop.

6.16 TO TEST ULTRASONIC Sensor

Load the ultasonic.hex file using programming tool FLASH MAGIC.

folder name: 16 Ultrasonic\Objects

path: SOFTWARES\16 Ultrasonic\Objects\ultasonic.hex

Result: After downloading the program, open the Terminal window from Tools Wizard in Flash Magic. Set the com port, baud rate as 9600 and press OK. Reset the trainer kit, the output of the Ultrasonic sensor will be displayed on the terminal as **"Ultrasonic sensor"** **"Distance = ***.**cm"** where ***.** is the output value of the sensor. This will be in continues loop.

6.17 TO TEST THERMAL Sensor

Load the Therm.hex file using programming tool FLASH MAGIC.

folder name: 17 Thermal

path: SOFTWARES\17 Thermal\Therm.hex

Result: After downloading the program, open the Terminal window from Tools Wizard in Flash Magic. Set the com port, baud rate as 9600 and press OK. Reset the trainer kit, the output of the Thermal sensor will be displayed on the terminal as

"Wel-Come to tmissystems.com, ADC and UART Configured @LPC2148

Serial Communication @ 9600 baudrate, 8 bits, no Parity, 1 Stop bit

Therm Voltage=1.74 V

Therm temp=27.562 Deg".

This will be in continues loop.

6.18 To Test Internal RTC Test:

Load the INT_RTC.hex file using programming tool FLASH MAGIC.

folder name: 18 INTERNAL_RTC

path: SOFTWARES\18 INTERNAL_RTC\INT_RTC.hex

Result: Internal RTC is operated based on the commands sent from PC terminal through UART0. A read and write operation is done and read values are sent to serial port UART0. A read operation will be continued.

1.RTC WRITE
2.RTC READ
PRESS 1 or 2

RTC DATA WRITTEN

1.RTC WRITE
2.RTC READ
PRESS 1 or 2

RTC DATA IS

Year	Month	DOY	DOW	DOM	HOUR	MIN	SEC
2014	12	365	6	31	23	59	59

Year	Month	DOY	DOW	DOM	HOUR	MIN	SEC
2015	01	1	7	01	00	00	00

Year	Month	DOY	DOW	DOM	HOUR	MIN	SEC
2015	01	1	7	01	00	00	01